Search Problems

CS5491: Artificial Intelligence

ZHICHAO LU

Content Credits: Prof. Wei's CS4486 Course
and Prof. Boddeti's AI Course

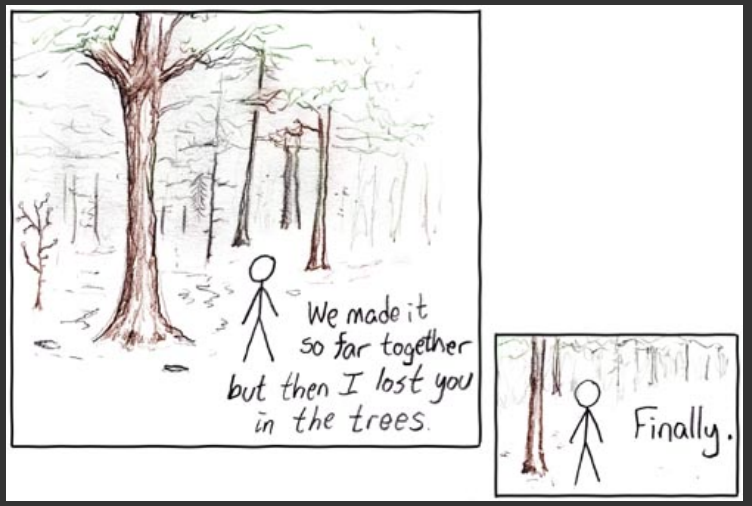
TODAY

Agents that Plan Ahead

Search Problems

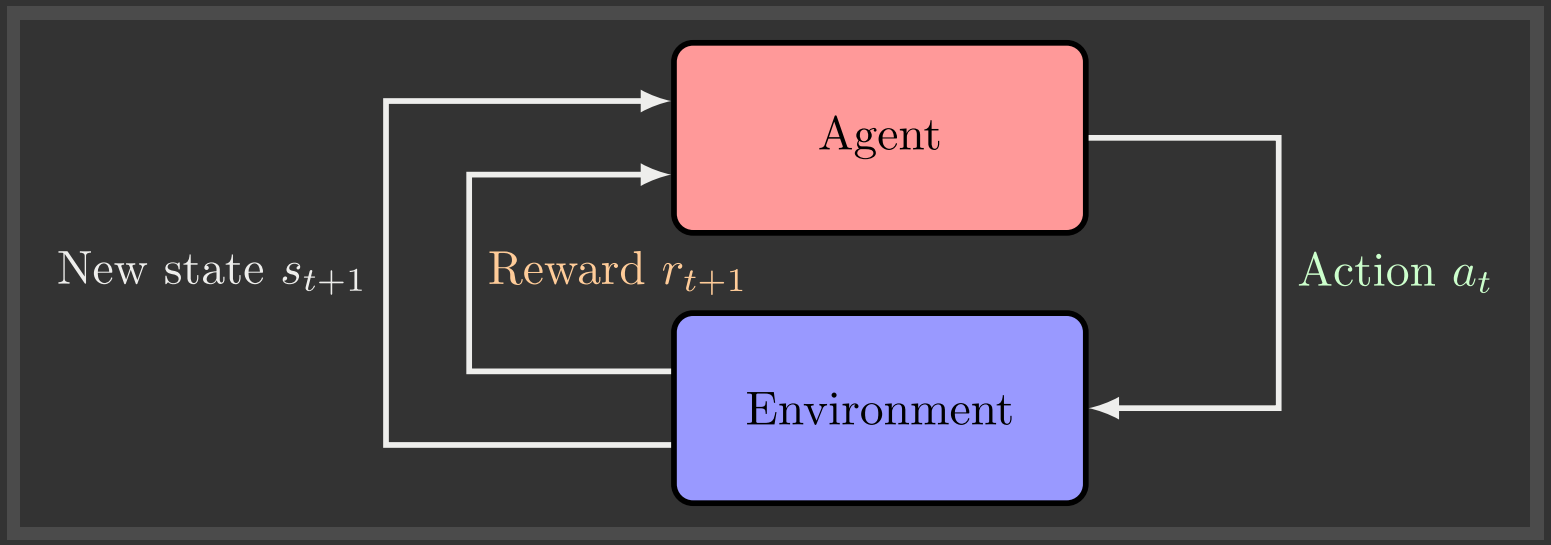
Reading

- Today's Lecture: RN Chapter 3.1-3.4
- Next Lecture: RN Chapter 3.1-3.4

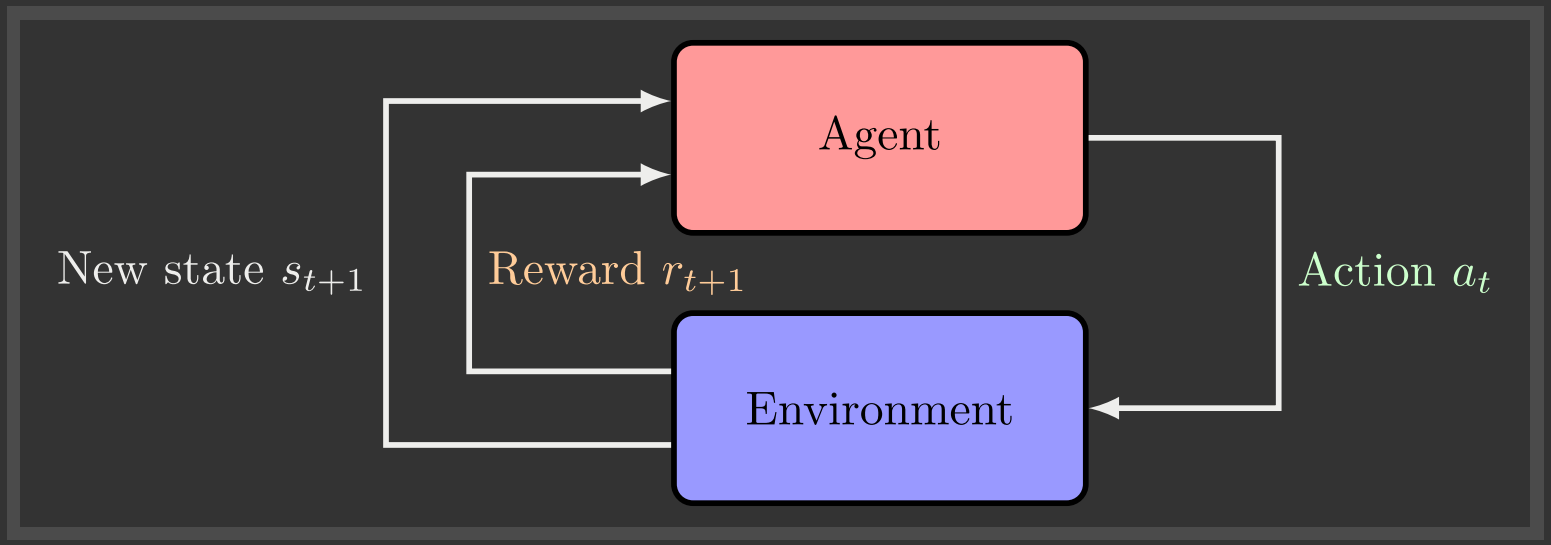


XKCD

REMINDER: RATIONAL AGENTS

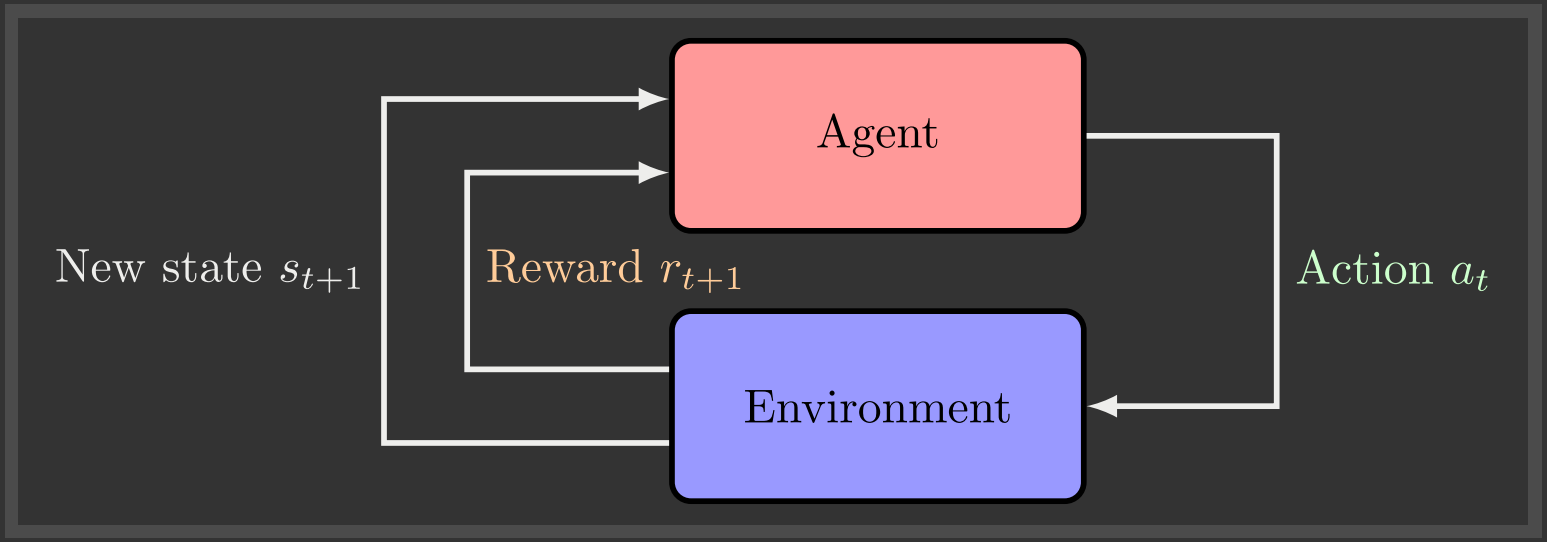


REMINDER: RATIONAL AGENTS



An **agent** is an entity that *perceives* and *acts*.

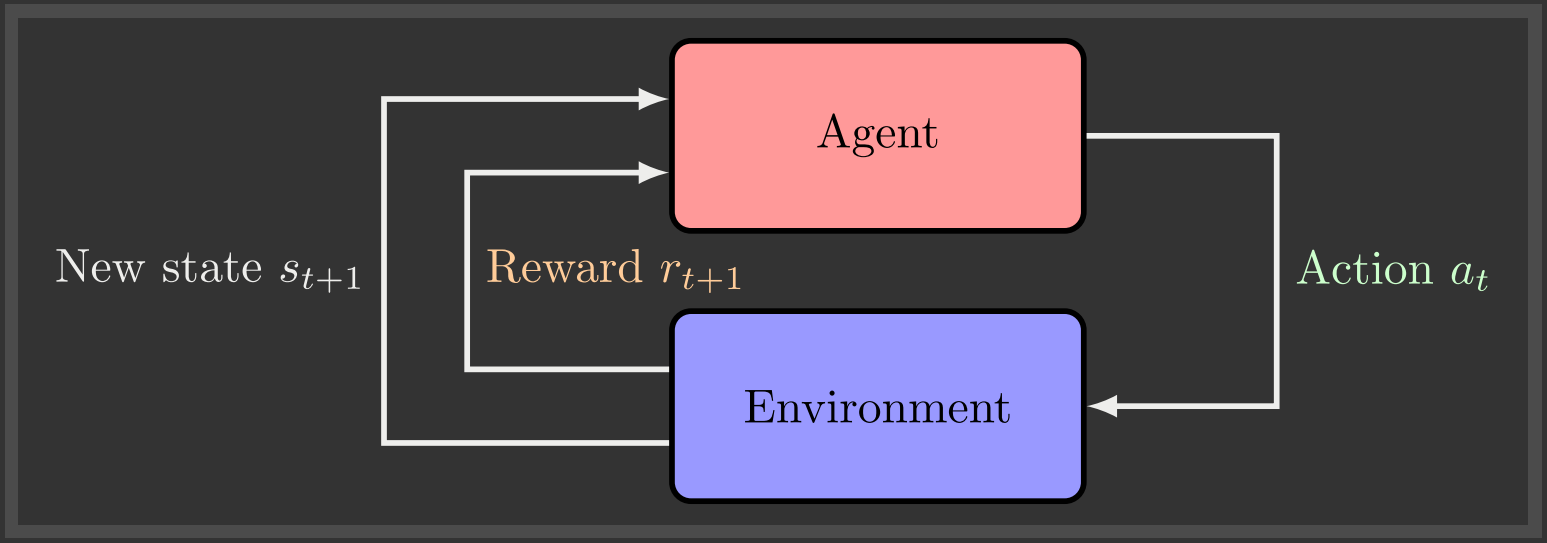
REMINDER: RATIONAL AGENTS



An **agent** is an entity that *perceives* and *acts*.

A **rational agent** selects actions that maximize its (expected) **utility**.

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An **agent** is an entity that *perceives* and *acts*.

A **rational agent** selects actions that maximize its (expected) **utility**.

Characteristics of the **percepts**, **environment**, and **action space** dictate techniques for selecting rational actions.

RATIONAL AGENTS



RATIONAL AGENTS

Are rational agents omniscient?



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- No - they are limited by what they can perceive



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Do rational agents explore and learn?

- Yes - essential qualities required in unknown environments

So, rational agents are not necessarily successful, but they are autonomous.



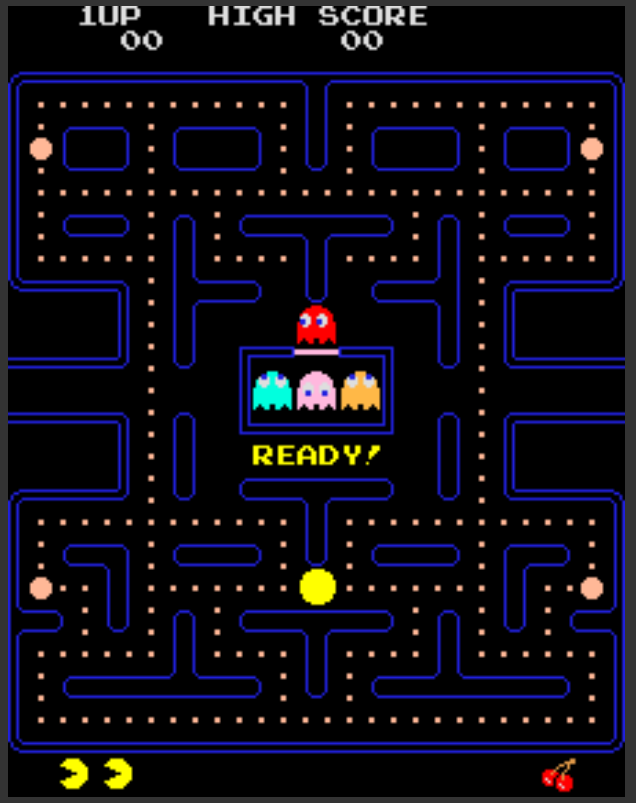
RATIONAL AGENTS

Performance Measure

Environment

Actuators

Sensors



RATIONAL AGENTS

Performance Measure

- -1 per time step;+10 food; +500 win; -500 die; +200 hit scared ghost

Environment

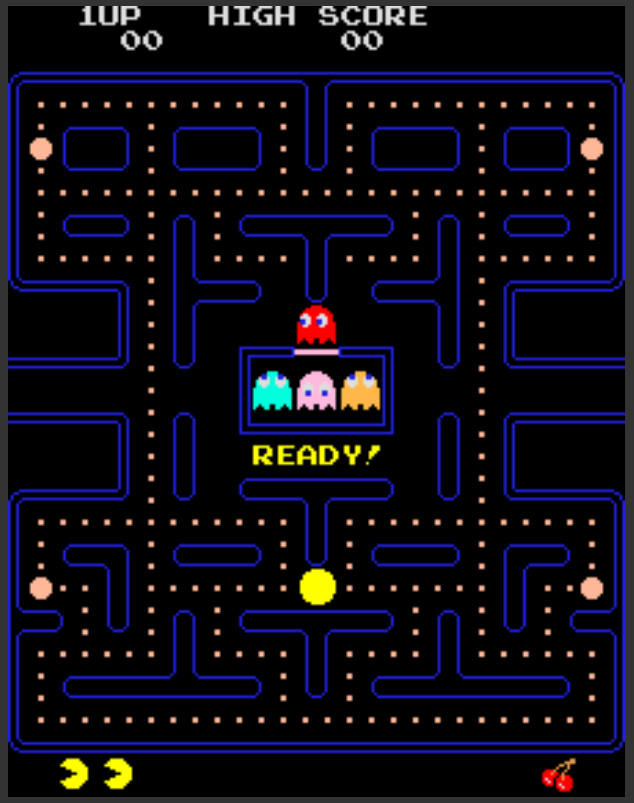
- Pacman dynamics + ghost behavior

Actuators

- North, West, East, South, Stop (idle)

Sensors

- entire state is visible



ROBOTAXI - PEAS

Performance Measure

Environment

Actuators

Sensors



ROBOTAXI - PEAS

Performance Measure

- income, happy customer, vehicle costs, fines, car, insurance

Environment

- streets, other drivers, customers, etc.

Actuators

- steering, brake, gas, display/speaker

Sensors

- cameras, lidar, radar, ultrasonic, accelerometer, mechanical sensors, microphone, etc.



ENVIRONMENT CATEGORIZATION

	Pacman	**Robotaxi**
Fully or Partially Observable		
Single or Multi Agent		
Deterministic or Stochastic		
Static or Dynamic		
Discrete or Continuous		



ENVIRONMENT CATEGORIZATION

	Pacman	**Robotaxi**
Fully or Partially Observable	fully	partial
Single or Multi Agent	multi	multi
Deterministic or Stochastic	deterministic	stochastic
Static or Dynamic	static	dynamic
Discrete or Continuous	discrete	continuous

ENVIRONMENT CATEGORIZATION

Reflex agents:

- Choose actions based on current observation (and maybe memory)
- May have memory or a model of the world's current state.
- Do not consider future consequences of actions
- Consider how the world IS as opposed to how it would be.

Can a reflex agent be rational?



AGENTS THAT PLAN AHEAD

Planning Agents:

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Planning Agents:

- Decision based on predicted consequences of actions



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Spectrum of Deliberativeness:

AGENTS THAT PLAN AHEAD

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Spectrum of Deliberativeness:

- Generate complete, optimal plan offline, then execute



AGENTS THAT PLAN AHEAD

Planning Agents:

- Decision based on predicted consequences of actions
- Must have a transition model : how the world evolves in responses to actions
- Must formulate a goal
- Consider how the world WOULD BE as opposed to how it is

Spectrum of Deliberativeness:

- Generate complete, optimal plan offline, then execute
- Generate simple, greedy plan, start executing, replan if necessary



SEARCH PROBLEMS



WHAT ARE SEARCH PROBLEMS?

A search problem consists of:



WHAT ARE SEARCH PROBLEMS?

A search problem consists of:

- A state space:



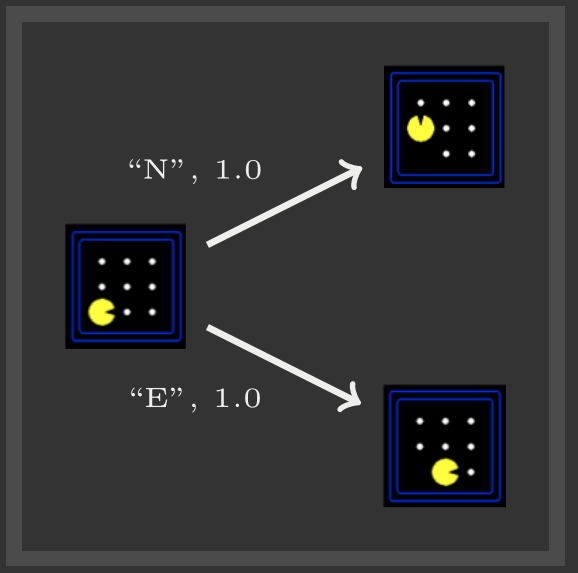
WHAT ARE SEARCH PROBLEMS?

A search problem consists of:

- A state space:



- A successor function (actions, costs):



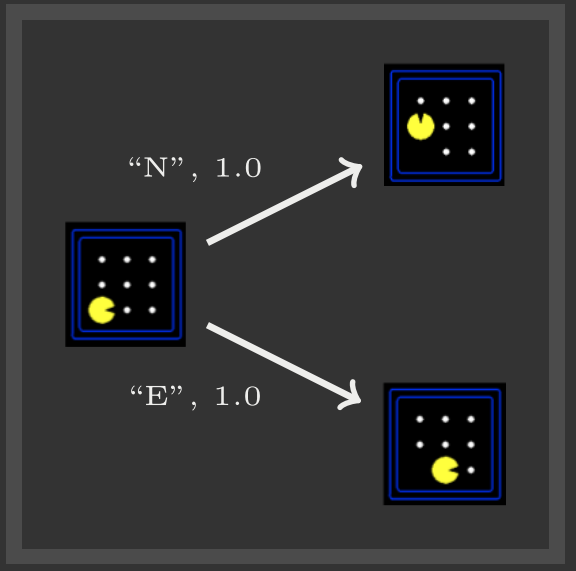
WHAT ARE SEARCH PROBLEMS?

A search problem consists of:

- A state space:



- A successor function (actions, costs):
- A start state and goal test



WHAT ARE SEARCH PROBLEMS?

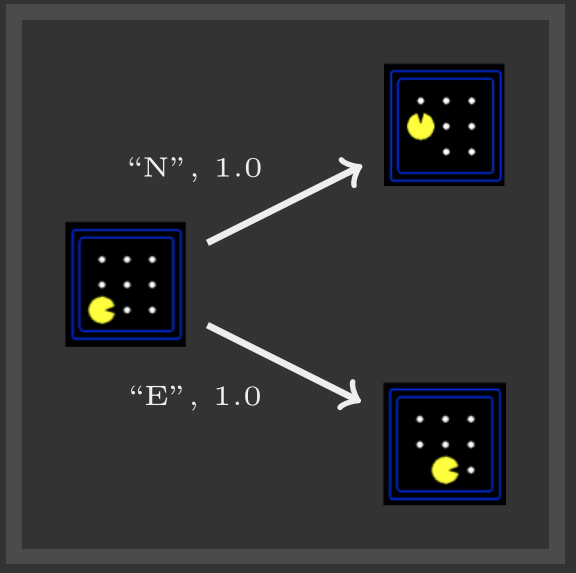
A search problem consists of:

- A state space:



- A successor function (actions, costs):
- A start state and goal test

A solution is a sequence of actions (a plan) which transforms the start state to the goal test.

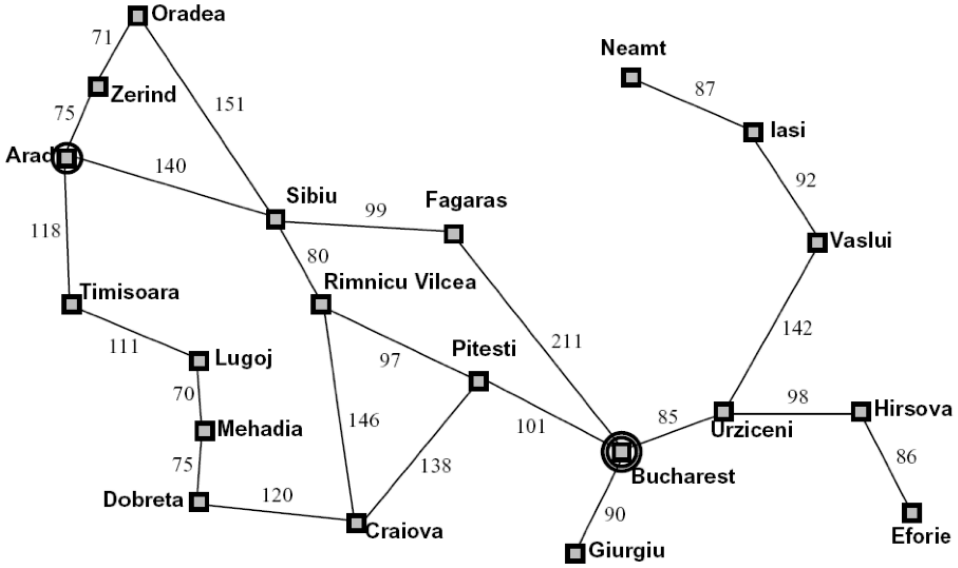


SEARCH PROBLEMS ARE MODELS

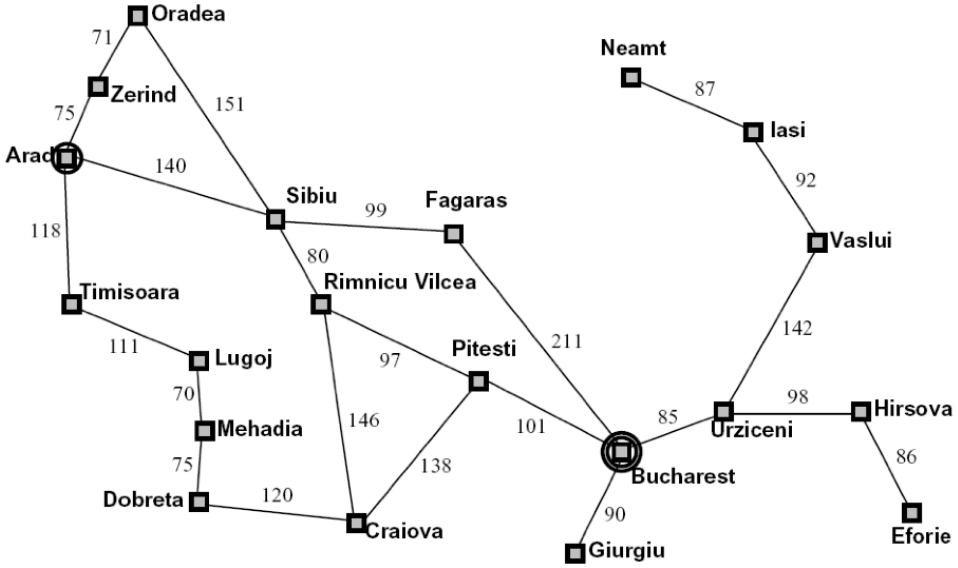




EXAMPLE: TRAVELING IN ROMANIA



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State Space:

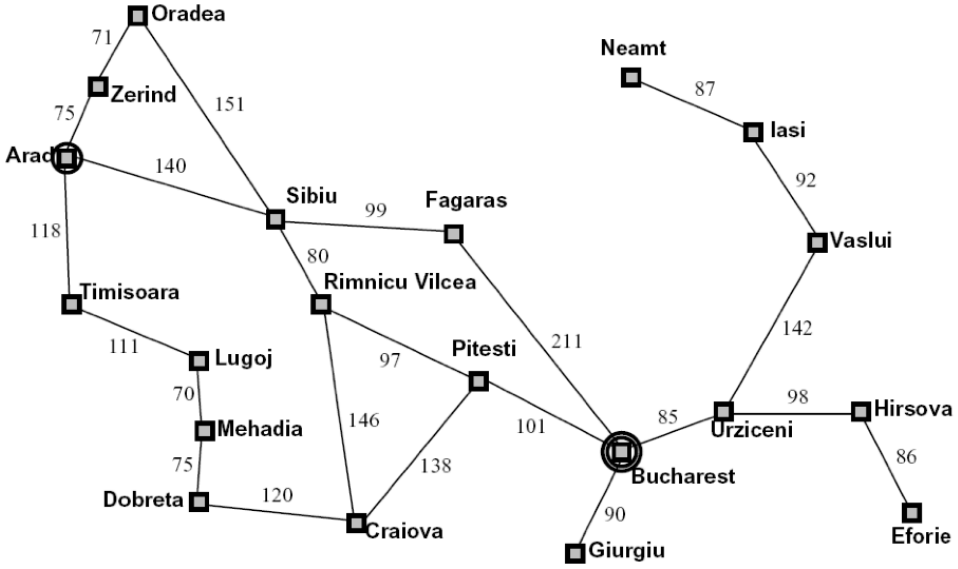
Successor Function:

Start Space:

Goal Test:

Solution?

EXAMPLE: TRAVELING IN ROMANIA



State Space:

- Cities

Successor Function:

- Roads: travel to adjacent city with cost=distance

Start Space:

- Arad

Goal Test:

- Is state == Bucharest?

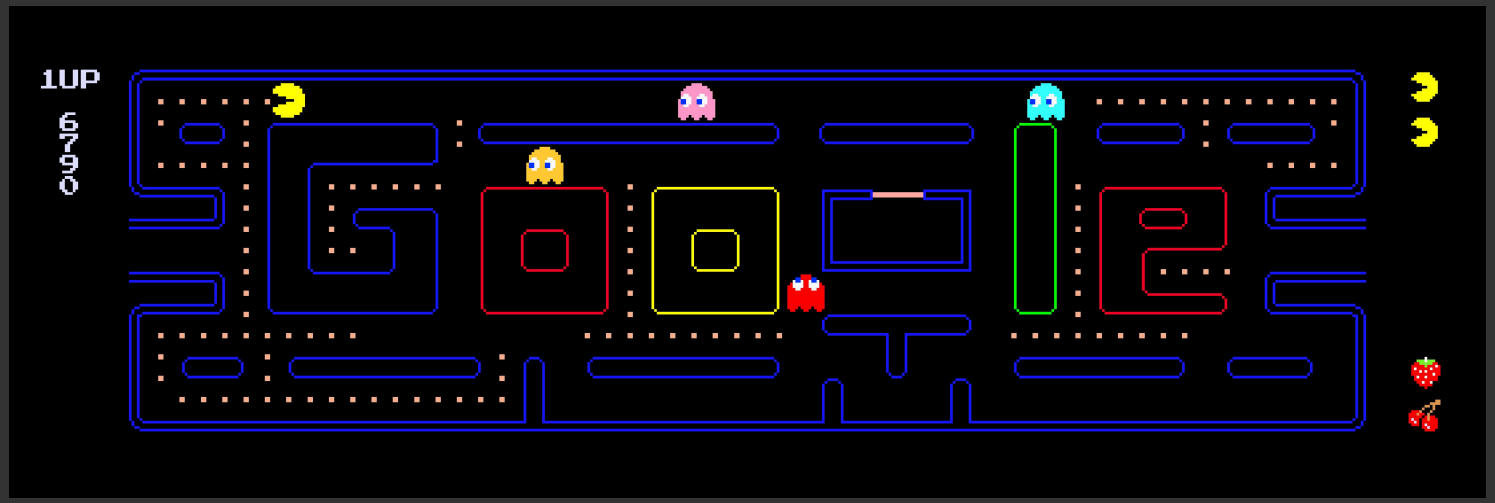
Solution?

WHAT IS IN A STATE SPACE?



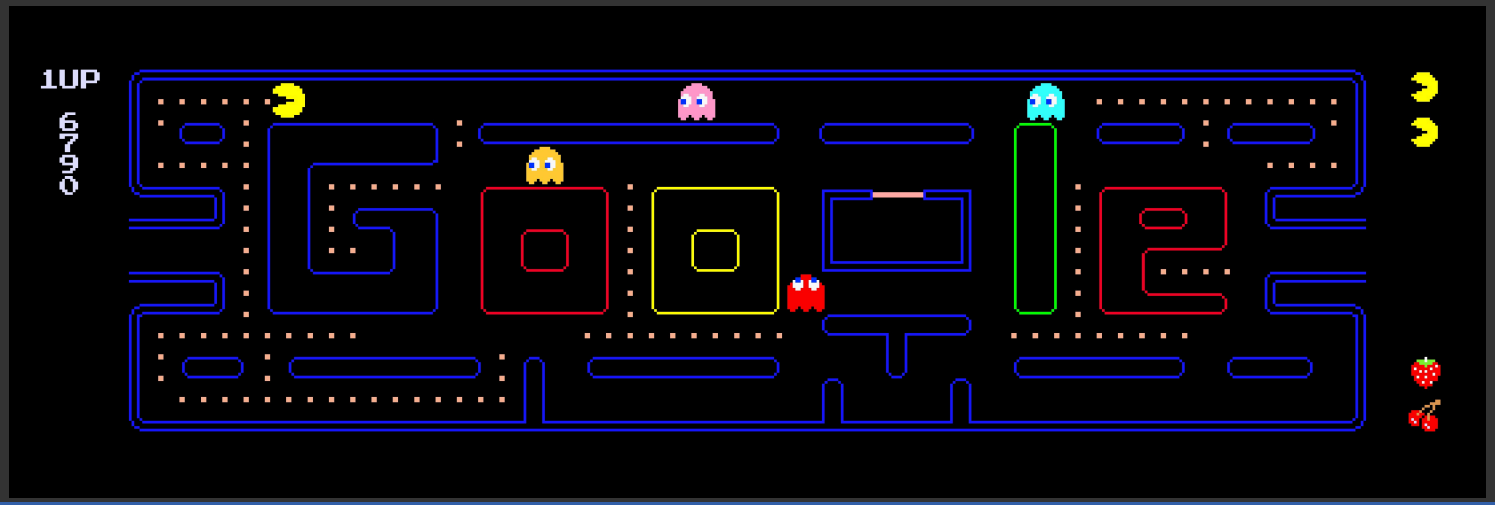
WHAT IS IN A STATE SPACE?

World State: Includes every last detail of the environment



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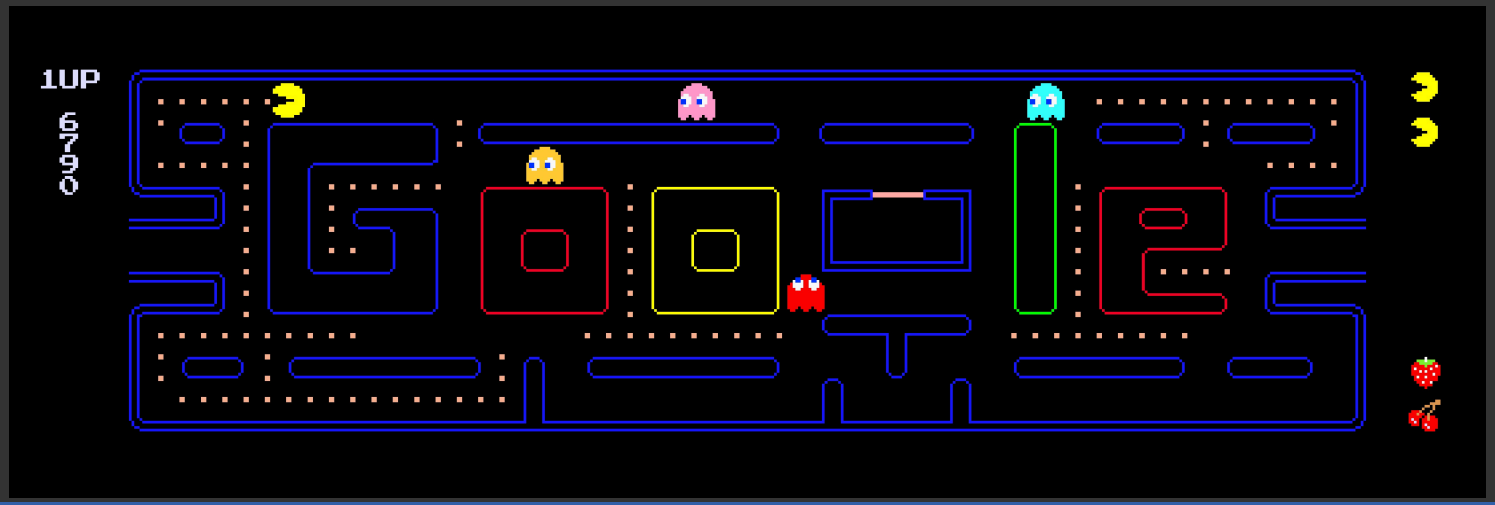
World State: Includes every last detail of the environment



Search State: Keeps only details necessary for planning

WHAT IS IN A STATE SPACE?

World State: Includes every last detail of the environment



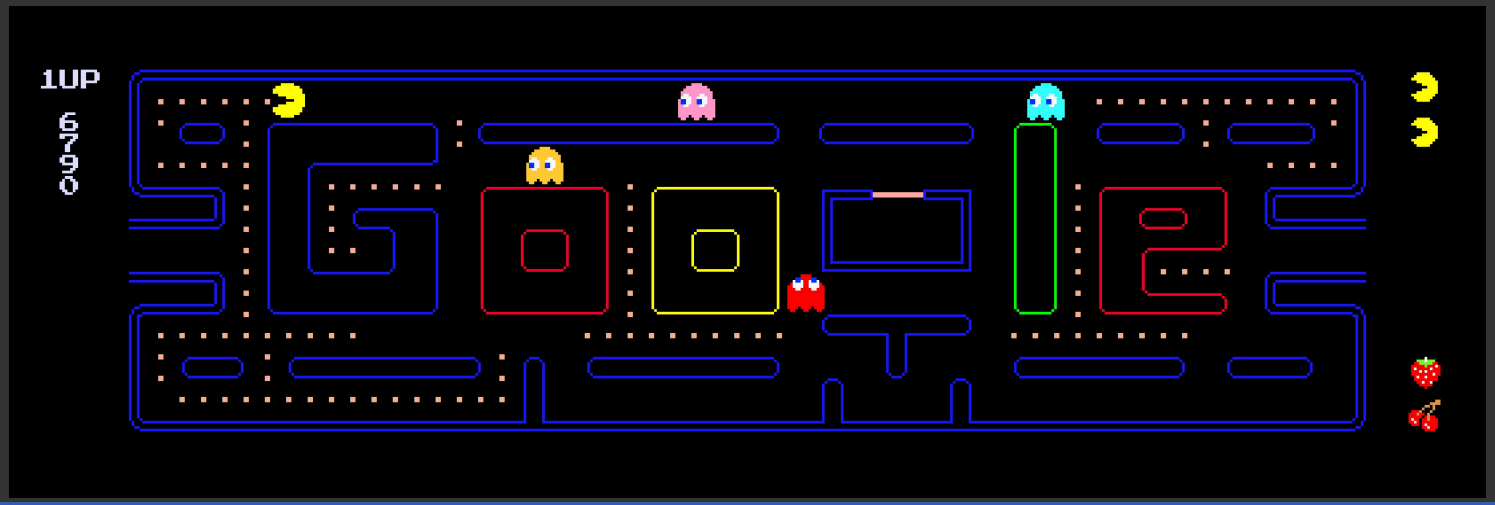
Search State: Keeps only details necessary for planning

Problem: Pathing

- States:
- Actions:
- Successor:
- Goal Test:

WHAT IS IN A STATE SPACE?

World State: Includes every last detail of the environment



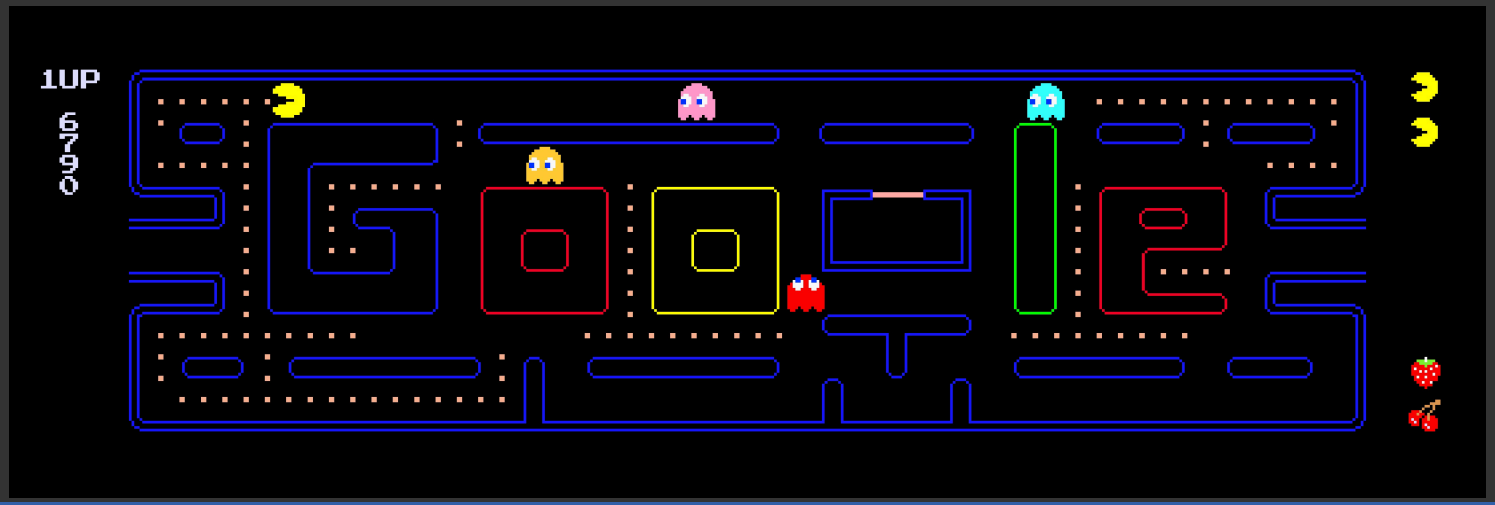
Search State: Keeps only details necessary for planning

Problem: Pathing

- States: (x,y) position
- Actions: NEWS
- Successor: update location
- Goal Test: Is (x,y) == END

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World State: Includes every last detail of the environment



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Problem: Pathing

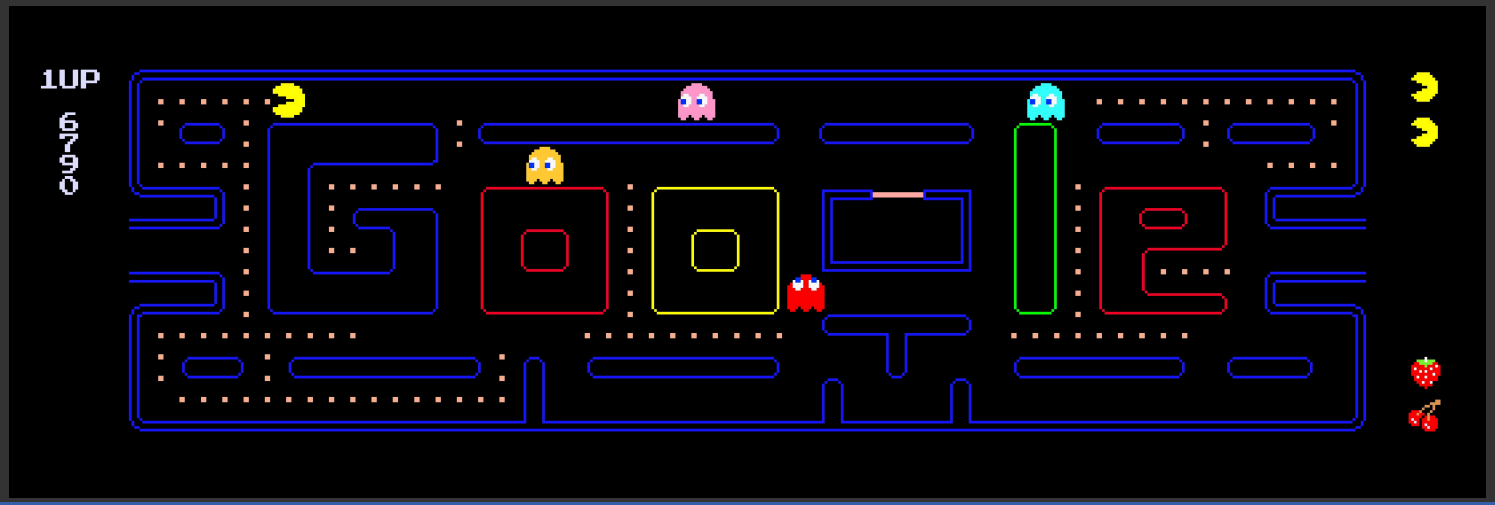
- States: (x,y) position
- Actions: NEWS
- Successor: update location
- Goal Test: Is (x,y) == END

Problem: Eat-All-Dots

- States:
- Actions:
- Successor:
- Goal Test:

WHAT IS IN A STATE SPACE?

World State: Includes every last detail of the environment



Search State: Keeps only details necessary for planning

Problem: Pathing

- States: (x,y) position
- Actions: NEWS
- Successor: update location
- Goal Test: Is (x,y) == END

Problem: Eat-All-Dots

- States: (x,y), boolean for each dot
- Actions: NEWS
- Successor: update location, boolean for dots
- Goal Test: All dot booleans are false

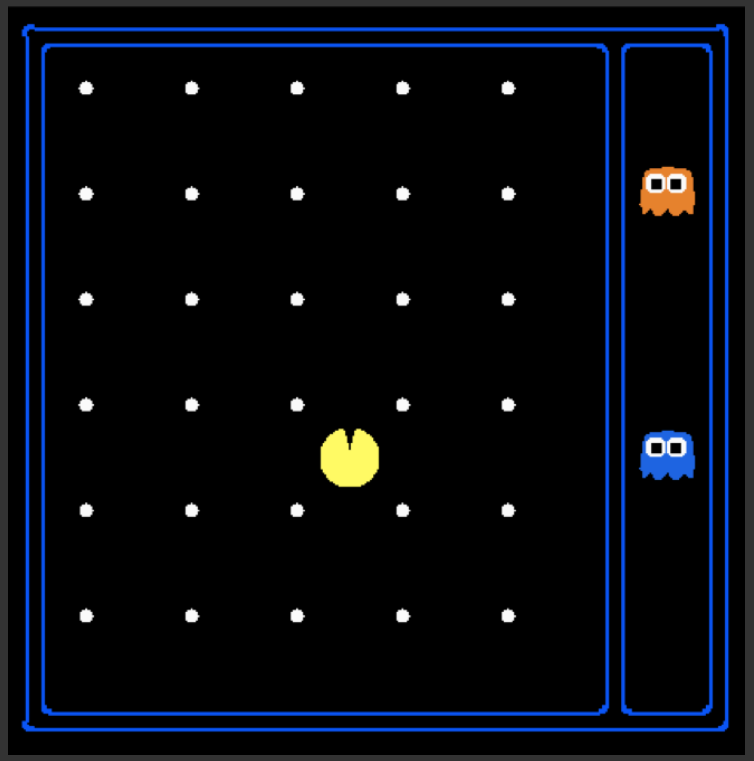
SEARCH SPACE SIZE

World State

- Agent Positions:
- Food Count:
- Ghost Positions:
- Agent Orientation:

Size

- World States:
- States for Pathing:
- States for Eat-All-Dots:



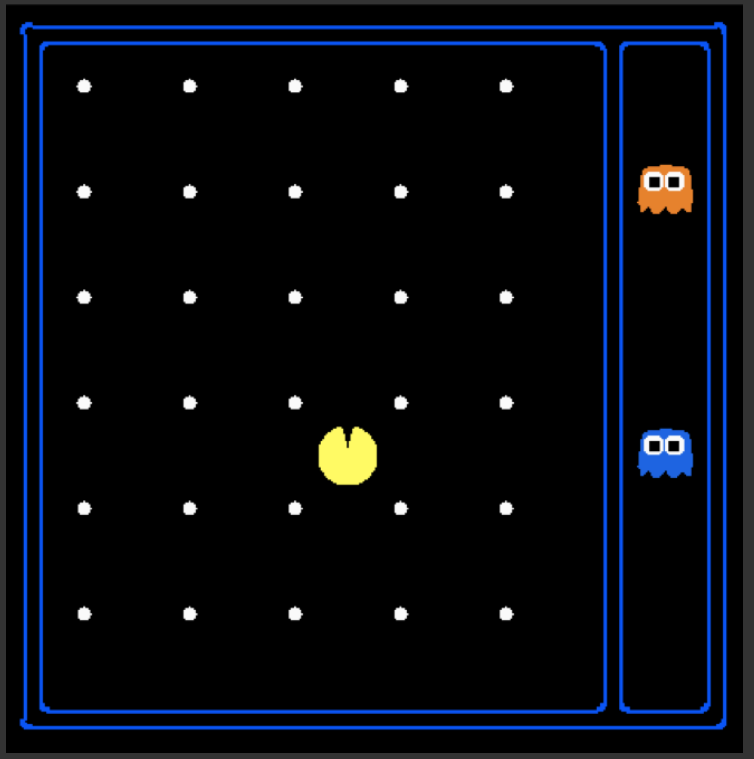
SEARCH SPACE SIZE

World State

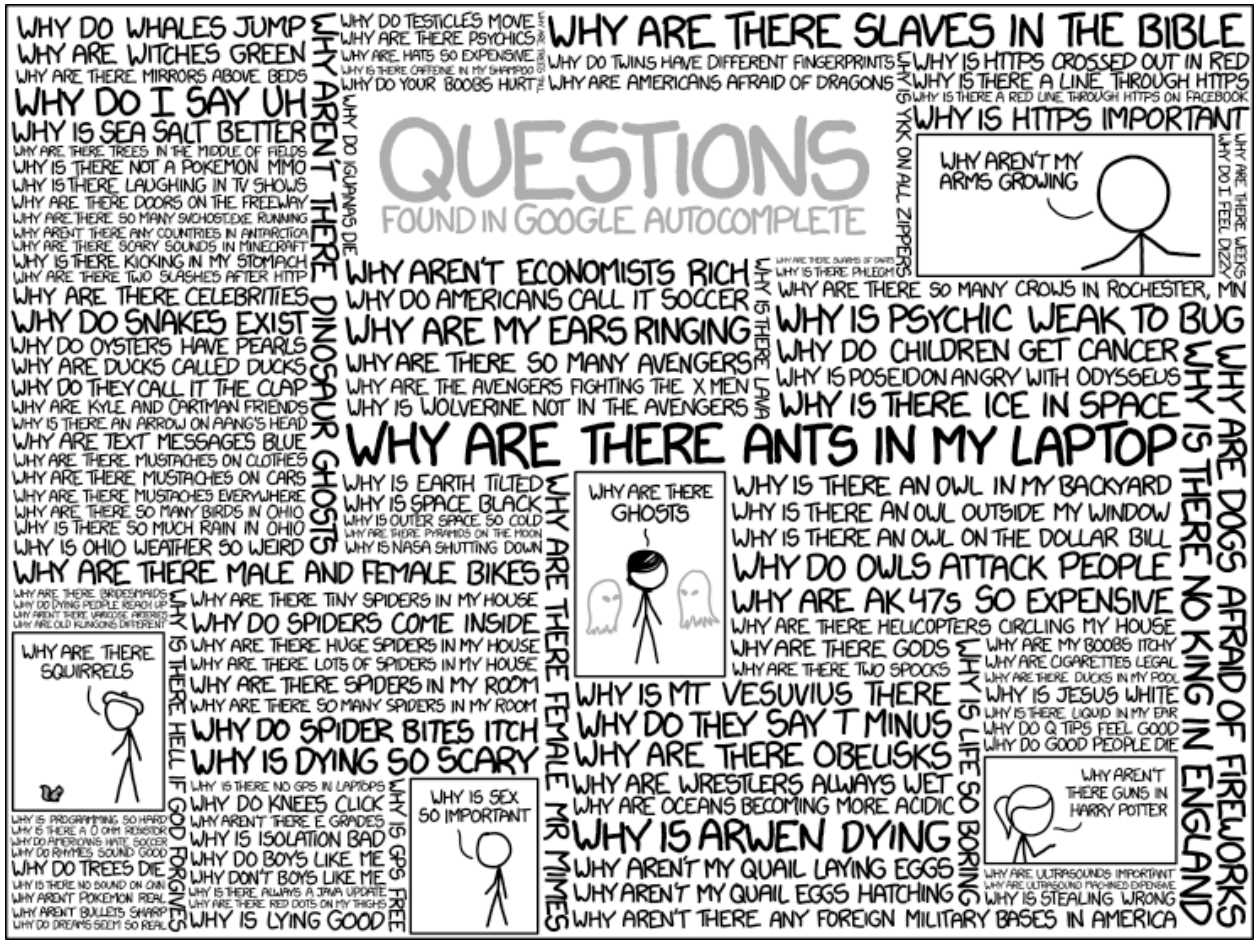
- Agent Positions: 120
- Food Count: 30
- Ghost Positions: 12
- Agent Orientation: NEWS

Size

- World States: $120 \times 2^{30} \times 12^2 \times 4$
- States for Pathing: 120
- States for Eat-All-Dots: 120×2^{30}



Q & A


QUESTIONS
FOUND IN GOOGLE AUTOCOMPLETE
WHY ARE THERE SLAVES IN THE BIBLE
WHY DO WHALES JUMP
WHY ARE WITCHES GREEN
WHY ARE THERE MIRRORS ABOVE BEDS
WHY DO I SAY UH
WHY IS SEA SALT BETTER
WHY ARE THERE TREES IN THE MIDDLE OF FIELDS
WHY IS THERE NOT A POKEMON MMO
WHY IS THERE LAUGHING IN TV SHOWS
WHY ARE THERE DOORS ON THE FREEWAY
WHY ARE THERE SO MANY SWCHOSTJDXE RUNNING
WHY ARENT THERE ANY COUNTRIES IN ANTARCTICA
WHY ARE THERE SCARY SOUNDS IN MINECRAFT
WHY IS THERE KICKING IN MY STOMACH
WHY ARE THERE TWO SLASHES AFTER HTTP
WHY ARE THERE CELEBRITIES
WHY DO SNAKES EXIST
WHY DO OYSTERS HAVE PEARLS
WHY ARE DUCKS CALLED DUCKS
WHY DO THEY CALL IT THE CLAP
WHY ARE KYLE AND CARTMAN FRIENDS
WHY IS THERE AN ARROW ON AANG'S HEAD
WHY ARE TEXT MESSAGES BLUE
WHY ARE THERE MUSTACHES ON CLOTHES
WHY ARE THERE MUSTACHES ON CARS
WHY ARE THERE MUSTACHES EVERYWHERE
WHY ARE THERE SO MANY BIRDS IN OHIO
WHY IS THERE SO MUCH RAIN IN OHIO
WHY IS OHIO WEATHER SO WEIRD
WHY ARE THERE MALE AND FEMALE BIKES
WHY ARE THERE BRIDESMAIDS
WHY DO DYING PEOPLE REACH UP
WHY AREN'T THERE VARICOSE AFFERETS
WHY ARE OLD KLINGONS DIFFERENT
WHY ARE THERE SQUIRRELS
WHY IS PROGRAMMING SO HARD
WHY IS THERE A 0 OHM RESISTOR
WHY DO AMERICANS HATE SOCCER
WHY DO RHYMES SOUND GOOD
WHY DO TREES DIE
WHY IS THERE NO SOUND ON OVA
WHY AREN'T POKEMON REAL
WHY AREN'T BULLETS SHARP
WHY DO DREAMS SEEM SO REAL
WHY ARE THERE TINY SPIDERS IN MY HOUSE
WHY DO SPIDERS COME INSIDE
WHY ARE THERE HUGE SPIDERS IN MY HOUSE
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE
WHY ARE THERE SPIDERS IN MY ROOM
WHY ARE THERE SO MANY SPIDERS IN MY ROOM
WHY DO SPIDER BITES ITCH
WHY IS DYING SO SCARY
WHY IS THERE NO GPS IN LAPTOPS
WHY DO KNEES CLICK
WHY AREN'T THERE E GRADES
WHY IS ISOLATION BAD
WHY DO BOYS LIKE ME
WHY DON'T BOYS LIKE ME
WHY IS THERE ALWAYS A JAVA UPDATE
WHY ARE THERE RED DOTS ON MY THIGHS
WHY IS LYING GOOD
WHY DO TESTICLES MOVE
WHY ARE THERE PSYCHICS
WHY ARE HATS SO EXPENSIVE
WHY IS THERE CAFFEINE IN MY SHAMPOO
WHY DO YOUR BOOBS HURT
WHY DO TWINS HAVE DIFFERENT FINGERPRINTS
WHY ARE AMERICANS AFRAID OF DRAGONS
WHY IS HTTPS CROSSED OUT IN RED
WHY IS THERE A LINE THROUGH HTTPS
WHY IS THERE A RED LINE THROUGH HTTPS ON FACEBOOK
WHY IS HTTPS IMPORTANT
WHY AREN'T MY ARMS GROWING
WHY ARE THERE SWANS AT DAWN
WHY IS THERE PHLEGM
WHY ARE THERE SO MANY CROWS IN ROCHESTER, MN
WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY ARE THERE SO MANY AVENGERS
WHY ARE THE AVENGERS FIGHTING THE X MEN
WHY IS WOLVERINE NOT IN THE AVENGERS
WHY IS PSYCHIC WEAK TO BUG
WHY DO CHILDREN GET CANCER
WHY IS POSEIDON ANGRY WITH ODYSSEUS
WHY IS THERE ICE IN SPACE
WHY ARE THERE ANTS IN MY LAPTOP
WHY IS EARTH TILTED
WHY IS SPACE BLACK
WHY IS OUTER SPACE SO COLD
WHY ARE THERE PYRAMIDS ON THE MOON
WHY IS NASA SHUTTING DOWN
WHY ARE THERE GHOSTS
WHY IS THERE AN OWL IN MY BACKYARD
WHY IS THERE AN OWL OUTSIDE MY WINDOW
WHY IS THERE AN OWL ON THE DOLLAR BILL
WHY DO OWLS ATTACK PEOPLE
WHY ARE AK 47s SO EXPENSIVE
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE
WHY ARE THERE GODS
WHY ARE THERE TWO SPOOKS
WHY IS MT VESUVIUS THERE
WHY DO THEY SAY T MINUS
WHY ARE THERE OBELISKS
WHY ARE WRESTLERS ALWAYS WET
WHY ARE OCEANS BECOMING MORE ACIDIC
WHY IS ARWEN DYING
WHY AREN'T MY QUAIL LAYING EGGS
WHY AREN'T MY QUAIL EGGS HATCHING
WHY AREN'T THERE ANY FOREIGN MILITARY BASES IN AMERICA
WHY ARE MY BOOBS ITCHY
WHY ARE CIGARETTES LEGAL
WHY ARE THERE DUCKS IN MY POOL
WHY IS JESUS WHITE
WHY IS THERE LIQUID IN MY EAR
WHY DO Q TIPS FEEL GOOD
WHY DO GOOD PEOPLE DIE
WHY AREN'T THERE GUNS IN HARRY POTTER
WHY ARE ULTRASOUNDS IMPORTANT
WHY ARE ULTRASOUND MACHINES EXPENSIVE
WHY IS STEALING WRONG
WHY IS SEX SO IMPORTANT
WHY ARE THERE FEMALE MR MIMES
WHY IS THERE HELL IF GOD FORGIVES
WHY IS GPS FREE
WHY IS LIFE SO BORING
WHY ARE THERE DINOSAUR GHOSTS
WHY AREN'T THERE
WHY DO IGUANAS DIE
WHY IS YKK ON ALL ZIPPERS
WHY ARE DOGS AFRAID OF FIREWORKS
WHY IS THERE NO KING IN ENGLAND
WHY ARE THERE WEEKS
WHY DO I FEEL DIZZY

XKCD